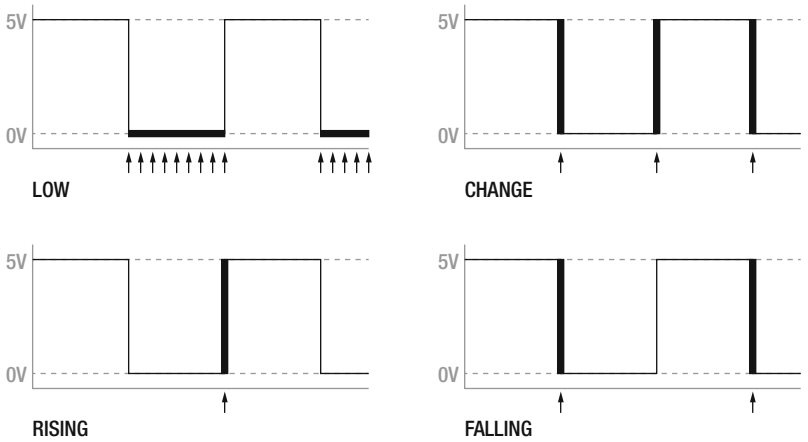


Figure 7-7. *State changes*

`LOW` triggers when the interrupt pin is in the `LOW` state. This mode will continuously trigger the interrupt for as long as it remains in this condition and because of this is not used as much as the other modes. `CHANGE` will trigger when the state of an interrupt pin changes either from `HIGH` to `LOW` or from `LOW` to `HIGH`. `RISING` will only trigger the interrupt when the signal goes from `LOW` to `HIGH` while `FALLING` will trigger on the reverse going from `HIGH` to `LOW`. Because of how our example circuit is connected, the button will send a signal of `HIGH` when triggered so we used the `RISING` mode to trigger the interrupt when the digital pin's state changes from `LOW` to `HIGH`.

Once the interrupt has been properly configured, often but not always in the `setup()` function, we need to write the ISR function. The function `selectHSB()` is a fairly good example of a short and succinct ISR. Its entire job is to increment the counter `i` and keep those values to a possible range of 0–3. Remember though, that in order for us to use the indexing variable `i` elsewhere in our sketch, we need to first use the `volatile` variable qualifier at the beginning of our sketch. Because the interrupt could sneak in there and change the value of a variable without the rest of the sketch knowing it, by using the `volatile` keyword we can tell the compiler to put the variable data in a more secure place in memory to prevent weird things from happening to our number.

detachInterrupt()

With our hardware interrupt enabled, it is possible that in a given application, we might need to change the mode of an interrupt, for example to change it from `RISING` to `FALLING`. To do this we would need to first stop the interrupt by using the `detachInterrupt()` function. Its syntax is fairly straightforward, as follows:

```
detachInterrupt(interrupt)
```

With only one parameter to determine which interrupt we are disabling, this parameter is specified as either 0 or 1. Once the interrupt has been disabled, we can then reconfigure it using a different mode in the `attachInterrupt()` function.